

# torch\_vn\_algebra: batched random operators, states, channels and constrained optimisation in finite-dimensional Type I von Neumann algebras

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## Abstract

`torch_vn_algebra` is a PyTorch library for numerical work in algebras  $\mathcal{M} = \bigoplus_{c=1}^C M_{k_c}(\mathbb{C})$ , the finite-dimensional Type I von Neumann algebras that describe quantum systems with superselection rules. Operators, states and channels are stored as batched, block-diagonal tensors, so that thousands of random samples, or thousands of independent starts of an optimisation, are processed at once on a CPU or a GPU. The library provides random operators with prescribed spectra and Haar-distributed eigenvectors, the three trace functionals that are natural on  $\mathcal{M}$ , states and Lüders conditioning, completely positive maps within and between sectors, plain and charge-conserving (fused) tensor products with partial traces, spin chains with a conserved magnetisation and momentum, sparse Lanczos and Krylov solvers for large sectors, Hamiltonian and Lindblad dynamics (including jumps between sectors), differentiable parametrisations with a prescribed Michelson contrast, and safeguards that estimate run time and memory before expensive steps. We describe the design, show how the library is used on short complete examples, and validate it against exact results from random matrix theory, quantum information and integrable spin chains (free-fermion spectra and dynamics, Heisenberg ring energies, Peschel’s entanglement formula, the GOE–Poisson crossover of disordered chains). Applied to the trace inequalities of our earlier work, constrained optimisation replaces Monte Carlo sampling: it recovers the sharp bounds that sampling misses in all but the smallest dimensions.

## Program summary

**Program title:** `torch_vn_algebra`

**Developer’s repository link:** [https://github.com/andre-a-no/von\\_neumann\\_type\\_i](https://github.com/andre-a-no/von_neumann_type_i)

**Licensing provisions:** MIT

**Programming language:** Python ( $\geq 3.9$ ), PyTorch ( $\geq 1.9$ , tested with 2.x)

**Nature of problem:** Numerical experiments in finite-dimensional Type I von Neumann algebras  $\bigoplus_c M_{k_c}(\mathbb{C})$ , which describe quantum systems with superselection rules or conserved charges: random operators with prescribed spectra, several trace functionals, states, channels within and between sectors, symmetric tensor products, spin chains, Hamiltonian and Lindblad dynamics, and extremal problems for trace inequalities under spectral constraints.

**Solution method:** Batched block-diagonal tensors processed by PyTorch linear algebra on CPUs and GPUs; exact Haar sampling; spectral functional calculus; Kraus representations; Lanczos and Krylov methods for sparse sector Hamiltonians; differentiable parametrisations with batched multistart gradient optimisation; run-time and memory estimates before expensive steps.

**Additional comments:** more than 250 unit tests, run in continuous integration; every table and figure of this paper is regenerated by `scripts/paper/run_all.sh`.

# 1 Introduction

**Where block-diagonal algebras arise.** Whenever a conserved quantity, a symmetry or a superselection rule splits a Hilbert space into sectors, the physically relevant operators are block diagonal. Their algebra is a finite-dimensional Type I von Neumann algebra

$$\mathcal{M} = \bigoplus_{c=1}^C M_{k_c}(\mathbb{C}) \quad \text{acting on} \quad \mathcal{H} = \bigoplus_{c=1}^C \mathbb{C}^{k_c}, \quad (1)$$

where  $C$  is the number of sectors (direct summands) and  $M_k(\mathbb{C})$  is the algebra of complex  $k \times k$  matrices. Every finite-dimensional von Neumann algebra is of this form [18], and its centre is spanned by the sector projections. The library also accepts nominal factor sizes  $n_c \geq k_c$ : this describes computations in the corner  $p\mathcal{M}_0p \cong \bigoplus_c M_{k_c}(\mathbb{C})$  of a larger algebra  $\mathcal{M}_0 = \bigoplus_c M_{n_c}(\mathbb{C})$ , with  $p = \bigoplus_c p_c$  and  $p_c$  a projection of rank  $k_c$ . The corner is again of the form (1); all operators, traces and normalisations refer to it, and in most applications  $k_c = n_c$ .

**From a symmetry to the algebra.** The link between physics and (1) is the commutant. If a group  $G$  (or a set of conserved quantities) acts unitarily on  $\mathcal{H}$  and  $\mathcal{H} \cong \bigoplus_\lambda V_\lambda \otimes \mathbb{C}^{m_\lambda}$  is its decomposition into irreducible representations  $V_\lambda$  with multiplicities  $m_\lambda$ , then the operators commuting with the action form the algebra  $\bigoplus_\lambda \mathbb{K}_{V_\lambda} \otimes M_{m_\lambda}(\mathbb{C}) \cong \bigoplus_\lambda M_{m_\lambda}(\mathbb{C})$ . For one conserved self-adjoint charge  $Q$  (particle number, magnetisation, Hamming weight) the irreducibles are one-dimensional and  $m_\lambda$  is the dimension of the eigenspace of  $Q$  with eigenvalue  $\lambda$ . A Hamiltonian that conserves  $Q$ , a time evolution generated by it, and every observable compatible with a superselection rule for  $Q$  lie in this commutant; the sectors are the eigenvalues of  $Q$ , and the centre consists of the functions of  $Q$ . Physical modelling with a symmetry is therefore computation in an algebra of the form (1).

Examples are abundant:

- superselection rules for charge, parity or particle number [19, 20] and the classical-quantum structure produced by decoherence and einselection [21];
- many-body Hamiltonians with conserved charges, whose exact diagonalisation and dynamics are carried out sector by sector [22, 23], and symmetry-resolved entanglement [24, 25];
- quantum information: decoherence-free subspaces and noiseless subsystems [26, 27], operator-algebra quantum error correction [28], and quantum algorithms that preserve a conserved quantity, such as the XY mixers of the quantum alternating operator ansatz, which keep the Hamming weight [29];
- random matrix theory, where symmetries decompose ensembles into blocks and spectral statistics are meaningful only inside a block [30, 12];
- machine learning: by Schur’s lemma the linear maps that commute with a group representation (equivariant layers) form exactly an algebra of the form (1), whose block sizes are the multiplicities of the irreducible representations [31, 32]; random-matrix methods are used to analyse trained weights [33];
- operator theory, our original motivation: inequalities between traces of products characterise the trace and the centre of an operator algebra [2, 3, 4, 5], and their failure can be quantified by spectral quantities such as the Michelson contrast [6, 7].

On  $\mathcal{M}$  the trace is not unique: every positive weighting of the sectors gives one, and statements such as “trace preserving” or “the dual map” depend on that choice as soon as maps change the sector structure.

**What existing software offers.** Table 1 compares the features relevant here. Dense and sparse linear algebra (NumPy/SciPy [34, 35], with ARPACK [36] for sparse eigenproblems) handles single matrices; QuTiP [8] and the quantum-information module of Qiskit [9] represent states, operators and channels on one Hilbert space; **dynamiqs** [10] adds batching, GPUs and automatic differentiation for Schrödinger and Lindblad dynamics; QuSpin [23] builds symmetry-reduced bases for exact diagonalisation; tensor-network libraries (ITensor [37], TeNPy [38]) store charge-conserving tensors in block-sparse form; RTNI [39] integrates Haar-random tensor networks symbolically. None of them treats the algebra (1) itself, with its centre, several traces, states and maps between algebras of different sector structure, as the basic object, and none combines it with batched sampling and optimisation over operators with prescribed spectra.

|                                                              | NumPy/<br>SciPy | QuTiP | Qiskit<br>q. info | dynamiqs | QuSpin | ITensor/<br>TeNPy | this<br>work |
|--------------------------------------------------------------|-----------------|-------|-------------------|----------|--------|-------------------|--------------|
| GPU execution                                                | –               | (✓)   | –                 | ✓        | –      | (✓)               | ✓            |
| batched over many operators                                  | (✓)             | –     | –                 | ✓        | –      | –                 | ✓            |
| automatic differentiation                                    | –               | (✓)   | –                 | ✓        | –      | (✓)               | ✓            |
| symmetry sectors of many-body systems                        | –               | –     | –                 | –        | ✓      | ✓                 | ✓            |
| algebra $\bigoplus_c M_{k_c}$ with centre and several traces | –               | –     | –                 | –        | –      | –                 | ✓            |
| channels within and between sectors                          | –               | (✓)   | (✓)               | –        | –      | –                 | ✓            |
| prescribed spectra, Haar/COE/CSE per sector                  | (✓)             | (✓)   | (✓)               | –        | –      | –                 | ✓            |
| optimisation under spectral constraints                      | –               | –     | –                 | –        | –      | –                 | ✓            |
| large single systems                                         | ✓               | ✓     | ✓                 | ✓        | ✓      | ✓                 | (✓)          |

Table 1: Features relevant to computations in the algebra (1), to the best of our knowledge at the time of writing (✓: supported; (✓): partly, through plug-ins, or with user code; –: not supported). The last row is where the specialised packages are the better choice.

**Why not one large matrix?** Nothing forces one to treat  $\mathcal{M}$  as a separate object: it embeds into  $M_N(\mathbb{C})$ ,  $N = \sum_c k_c$ , as the block-diagonal matrices, and every package above handles a single  $N \times N$  matrix (the case  $C = 1$  of (1)). The embedding is exact for linear algebra, so the question is only what it costs and what has to be supplied by hand.

- *Cost.* A product, an inverse or a spectral decomposition in  $M_N$  costs  $O(N^3)$  time and  $O(N^2)$  memory, in  $\mathcal{M}$  it costs  $\sum_c k_c^3$  and  $\sum_c k_c^2$ . For  $C$  equal blocks this is a factor  $C^2$  in time and  $C$  in memory; for the half-filling sector of a chain of  $L = 16$  spins ( $N = 2^{16}$ , largest block 12870) the dense  $M_N$  computation is out of reach, the block computation is routine.
- *Measures.* The unitary group of  $\mathcal{M}$  is  $\prod_c U(k_c)$ , a subgroup of  $U(N)$  of measure zero. A Haar-random element of  $U(N)$  is not in  $\mathcal{M}$ , and a random element of  $\mathcal{M}$  has to be produced block by block. The same holds for random states, for the circular ensembles and for any optimisation: an unconstrained gradient step in  $M_N$  leaves  $\mathcal{M}$ .
- *Traces.* On  $M_N$  the trace is unique up to a factor; on  $\mathcal{M}$  the faithful traces form a  $C$ -parameter family  $\sum_c w_c \text{Tr}_c$ . Densities, “trace preserving” and the dual of a map between algebras with different sectors depend on the choice of the weights; on the embedded matrices this information is no longer visible.
- *Centre.* The centre of  $\mathcal{M}$  (the sector projections) is what physicists call the classical or superselected part of the system; conditioning on a sector, post-selection and sector-resolved quantities are operations with the centre, and they have to be reconstructed from the block indices when  $\mathcal{M}$  is stored as one matrix.

The library keeps these four pieces of structure inside the data type. For  $C = 1$  it reduces to batched linear algebra in  $M_n(\mathbb{C})$ , and all its functions accept that case. The first version of this library [1] shows that the bookkeeping is error-prone: its complex Haar sampler was not Haar distributed, its circular ensembles were not the circular ensembles, and several conclusions drawn from Monte Carlo samples were wrong; all of this is corrected and tested here.

**Challenges.** Turning the algebra into an efficient and reliable tool raises concrete problems, each addressed by a design decision described in Section 3: (i) heterogeneous block sizes have to be mapped onto batched GPU kernels; (ii) trace conventions and dual maps must stay consistent when maps change the sectors; (iii) random ensembles must be exactly Haar (or COE/CSE) in every block; (iv) gradients propagated through eigen- or singular-vector decompositions (as in  $f(A)$  or  $|X|$ ) are numerically singular at degenerate spectra, exactly where extremal operators tend to lie; (v) extremal values sit on sets of tiny measure that random sampling does not reach in higher dimensions (Section 6); (vi) batched workloads easily exceed the time or memory budget of a device; (vii) the number field (real or complex) and the precision change both cost and results (Section 3).

**Objective and scope.** `torch_vn_algebra` is a library for numerical experiments *in* the algebra (1): sampling and optimising operators, states and channels, evaluating trace and spectral functionals, and formulating physical models (spin chains, open systems, noisy symmetric quantum circuits) whose symmetry is encoded in the sector structure. It is not a circuit simulator and not a tensor-network code, and for single large systems the specialised packages above remain the better choice; Section 8 quantifies where the library gains and where it does not.

**Contributions and outline.** Section 2 fixes notation and conventions. Section 3 describes the software stack, the workflow, how the challenges (i)–(vii) are met, and the choice of number field and precision. Section 4 shows the library on short complete programs, including a noisy quantum circuit. Section 5 validates it against exact results and checks the trace functionals. Section 6 replaces Monte Carlo sampling by constrained optimisation for the trace inequalities of [1]. Section 7 covers charge-conserving tensor products and spin chains, including momentum sectors, sparse Krylov solvers and particle loss. Section 8 compares with NumPy/SciPy, Section 9 reports performance, and Section 10 discusses limitations.

## 2 Mathematical setting

**Algebra and traces.** Let  $\mathcal{M} = \bigoplus_{c=1}^C M_{k_c}(\mathbb{C})$  act on  $\mathcal{H} = \bigoplus_c \mathbb{C}^{k_c}$ ;  $A \in \mathcal{M}$  is block diagonal,  $A = \bigoplus_c A_c$ . Every trace on  $\mathcal{M}$  is  $\text{tr}_w(A) = \sum_c w_c \text{Tr } A_c$  with weights  $w_c \geq 0$ , and it is faithful iff all  $w_c > 0$ . The library implements

$$\text{Tr}_{\text{blunt}} : w_c = 1, \quad \text{Tr}_{\text{norm}} : w_c = \frac{1}{k_c}, \quad \tau_{\text{vN}} : w_c = \frac{1}{C k_c},$$

where  $\tau_{\text{vN}}$  is the faithful normal tracial state that gives every sector the same weight. For  $A \in \mathcal{M}$ ,  $|A| = (A^* A)^{1/2}$ ; for  $A \geq 0$  the Michelson contrast is  $\Delta(A) = (\lambda_{\max} - \lambda_{\min})/(\lambda_{\max} + \lambda_{\min}) \in [0, 1]$ , with the extreme eigenvalues taken over all sectors;  $\Delta(A) = 0$  if and only if  $A$  is a multiple of the identity.

**States.** A state  $\omega$  on  $\mathcal{M}$  is represented by its density  $\rho \geq 0$  with respect to  $\text{Tr}_{\text{blunt}}$ ,  $\omega(A) = \text{Tr}(\rho A)$ ,  $\sum_c \text{Tr } \rho_c = 1$ . The restriction of  $\omega$  to the centre is the classical distribution  $p_c = \text{Tr } \rho_c$  over sectors. The density of the same state with respect to  $\text{tr}_w$  is  $\rho_c^{(w)} = \rho_c / w_c$ ; the library converts between the

conventions explicitly, because statements such as “trace preserving” or “the dual map” depend on them as soon as the sectors change.

**Maps between sectors.** Let  $\mathcal{M}_{\text{in}} = \bigoplus_c M_{k_c}$  and  $\mathcal{M}_{\text{out}} = \bigoplus_d M_{m_d}$ . Every normal completely positive map  $\Phi : \mathcal{M}_{\text{in}} \rightarrow \mathcal{M}_{\text{out}}$  (in the Schrödinger picture) is a sum of completely positive maps between blocks,

$$\Phi(\rho)_d = \sum_c \sum_i K_i^{dc} \rho_c (K_i^{dc})^*, \quad K_i^{dc} : \mathbb{C}^{k_c} \rightarrow \mathbb{C}^{m_d}.$$

It preserves  $\text{Tr}_{\text{blunt}}$  iff  $\sum_{d,i} (K_i^{dc})^* K_i^{dc} = 1_c$  for all  $c$ , and  $T_{dc} = \text{Tr} \Phi_{dc}(\rho_c) / \text{Tr} \rho_c$  are the transition probabilities between sectors for the input state  $\rho$  (independent of  $\rho$  iff every  $\sum_i (K_i^{dc})^* K_i^{dc}$  is a multiple of  $1_c$ ; by default the library uses the maximally mixed state of each input sector). The dual with respect to  $\text{tr}_w$  on both sides is  $\Phi^{*,w}(A)_c = \sum_{d,i} (w_d^{\text{out}} / w_c^{\text{in}}) (K_i^{dc})^* A_d K_i^{dc}$ ; for sector-preserving maps on one algebra the weights cancel and all three traces give the same dual.

**Tensor products and fusion of sectors.**  $(\bigoplus_c M_{k_c}) \otimes (\bigoplus_d M_{m_d}) = \bigoplus_{(c,d)} M_{k_c m_d}$  is again of Type I with sectors labelled by pairs. If the sectors carry conserved charges  $q_c, q'_d$  (particle numbers, magnetisation), the operators of the composite system that conserve only the *total* charge form the larger algebra

$$\mathcal{M}_1 \hat{\otimes} \mathcal{M}_2 = \bigoplus_q M_{n_q}, \quad n_q = \sum_{q_c + q'_d = q} k_c m_d,$$

in which all pairs with the same total charge are merged into one sector. It contains  $\mathcal{M}_1 \otimes \mathcal{M}_2$  (block-diagonally in every merged sector), but also operators such as  $S_i^+ S_j^-$  that move charge between the factors. For additive charges, and distinct charges on distinct sectors of  $\mathcal{M}_1$ , the partial trace  $(\text{Tr}_2 X)_c = \sum_d \text{Tr}_{m_d} X_{(c,d),(c,d)}$  still lands in  $\mathcal{M}_1$ : a block between pairs  $(c,d) \neq (c',d)$  with equal total charge would need  $q_c = q_{c'}$ , hence  $c = c'$ . In both cases the partial trace is a map between sectors in the sense above, and its dual with respect to  $\text{Tr}_{\text{blunt}}$  is the embedding  $A \mapsto A \otimes 1$ .

**Spin chains.** For  $L$  spins  $1/2$  with conserved  $S_{\text{tot}}^z$ , iterating the fused product of single-site algebras  $M_1 \oplus M_1$  (charges  $0, 1$ ) gives  $\mathcal{M} = \bigoplus_{N=0}^L M_{\binom{L}{N}}(\mathbb{C})$ ,  $N$  the number of up spins, which contains every  $S_{\text{tot}}^z$ -conserving Hamiltonian, e.g. the XXZ chain  $H = \sum_{\langle ij \rangle} [\frac{J}{2} (S_i^+ S_j^- + S_i^- S_j^+) + J \Delta S_i^z S_j^z] + \sum_i h_i S_i^z$ .

**Dynamics.** For  $H = H^* \in \mathcal{M}$  and jump operators  $L_j \in \mathcal{M}$  the Gorini–Kossakowski–Sudarshan–Lindblad equation  $\dot{\rho} = -i[H, \rho] + \sum_j \gamma_j (L_j \rho L_j^* - \frac{1}{2} \{L_j^* L_j, \rho\})$  preserves every sector, so the sector probabilities  $p_c$  are conserved (superselection). Jumps between sectors, such as particle loss, are described by maps between sectors as above (Section 7).

### 3 Design and implementation

**Software stack.** The library is pure Python on top of PyTorch [40] ( $\geq 1.9$ ; tested with 2.x) and NumPy [34]. All dense linear algebra goes through `torch.linalg`, i.e. LAPACK (MKL or OpenBLAS) on CPUs and cuSOLVER, cuBLAS and MAGMA on NVIDIA GPUs; sparse products use PyTorch’s CSR tensors. Automatic differentiation is PyTorch’s reverse mode. The validation and paper scripts additionally use SciPy, pandas and Matplotlib. Packaging follows `pyproject.toml`; the test suite runs in continuous integration on Python 3.10 and 3.12 (CPU), and the paper scripts record the PyTorch, CUDA and cuDNN versions, the device and its compute capability in every generated table.

**Workflow.** A computation proceeds in four steps. (1) An algebra is declared by its sector sizes, optionally sector charges, number field, precision and device (`TypeIAlgebra`, or a model such as `SpinChain` that creates it). (2) Objects are created in it: random operators with prescribed spectra, states, channels, Hamiltonians, or differentiable parametrisations. (3) They are combined lazily (products, functional calculus, channel application, tensor products, time evolution). (4) Functionals are evaluated for the whole batch at once (traces, spectra, contrasts, entropies, sector probabilities) and reduced to statistics, or fed into an optimiser that updates the parameters of step (2). Every step keeps the batch dimension, so the same code serves one operator and  $10^5$  samples.

| challenge                                       | solution in the library                                                                                                                                                 |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (i) blocks of different size on batched kernels | padded $(B, C, k_{\max}, k_{\max})$ layout, one batched call per sector; single-sector algebras and sparse sector Hamiltonians where padding would dominate             |
| (ii) conventions between sectors                | states stored as $\text{Tr}_{\text{blunt}}$ -densities with explicit conversions; duals of maps between sectors computed with the weights of the chosen trace           |
| (iii) exact random ensembles                    | Mezzadri's QR algorithm with phase correction, $\det^{-1/n}$ for $SU(n)$ , $W^T W$ and $W^R W$ for COE/CSE, validated against Weingarten moments and spacing statistics |
| (iv) singular gradients at degenerate spectra   | parametrisations that build operators from their spectra (no eigendecomposition in the forward pass), exact $ X $ from the parametrisation                              |
| (v) extremes missed by sampling                 | batched multistart optimisation with exact contrast constraints                                                                                                         |
| (vi) time and memory budgets                    | probing of batched kernels with an ETA warning, memory estimates checked against the free device memory, ETA for time stepping                                          |
| (vii) field and precision                       | two switches per algebra, conversions between them, measured trade-offs                                                                                                 |

Table 2: Implementation challenges (Section 1) and how they are addressed.

**Data layout.** An operator is a tensor of shape  $(B, C, k_{\max}, k_{\max})$ :  $B$  samples,  $C$  sectors, block  $c$  in the top-left  $k_c \times k_c$  corner and zeros elsewhere. Linear algebra is applied to the batch of active blocks of each sector, so one call to a batched LAPACK/cuSOLVER routine serves all samples. The padding costs memory when the sector sizes differ strongly (for a chain of 14 spins, 77 % of a padded operator is zeros); the library warns in that case, and `SpinChain` lets one work in single sectors  $M_{(N)}^{(L)}$  without padding, which is what dense diagonalisation needs anyway.

**Objects.** `Operator` (with `DensityMatrix` as a subclass for states), `Channel` (sector preserving) and `InterSectorChannel` (between sectors or algebras) are thin wrappers around such tensors; Table 3 lists the modules. Operations on operators are lazy: they record how to compute the result and do so on first use; the result is cached and the recipe dropped, so intermediates of long expressions can be freed. Laziness defers and skips work but does not fuse operations; where this matters the library uses direct formulas, e.g.  $\text{Tr}(\rho A) = \sum_{ij} \rho_{ij} A_{ji}$  in  $O(k^2)$  instead of forming  $\rho A$ .

**Random unitaries.** Haar unitaries on  $U(n)$  and  $O(n)$  are obtained from the QR decomposition of a Ginibre matrix with i.i.d. complex (respectively real) Gaussian entries after fixing the phases (signs) of  $\text{diag } R$  [11];  $SU(n)$  and  $SO(n)$  by multiplying with  $\det^{-1/n}$  or by a sign flip of one column; the circular orthogonal and symplectic ensembles as  $W^T W$  and  $W^R W$  with  $W$  Haar on  $U(n)$  ( $n$  even for  $W^R W$ ,  $W^R$  the quaternion dual) [12]. Section 5 checks all of them.

**Differentiation.** All operations are composed of differentiable PyTorch primitives, so gradients with respect to the matrix entries are available; functionals of the eigenvalues alone differentiate without problems, but gradients propagated through eigen- or singular vectors contain fac-

| module           | content                                                                                                                                                   |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>algebra</b>   | <code>TypeIAlgebra</code> , <code>Operator</code> : random operators, unitary ensembles, traces, functional calculus                                      |
| <b>states</b>    | <code>DensityMatrix</code> , random states, Gibbs states, Lüders rule, entropies, fidelity                                                                |
| <b>channels</b>  | <code>Channel</code> , <code>InterSectorChannel</code> , Choi matrices, standard and random channels                                                      |
| <b>composite</b> | plain and fused (charge-conserving) tensor products, <code>kron</code> , partial traces                                                                   |
| <b>chains</b>    | <code>SpinChain</code> : $U(1)$ and momentum sectors, XXZ Hamiltonians with batched disorder, sparse form, reduced states, jump operators between sectors |
| <b>krylov</b>    | sparse sector Hamiltonians, batched Lanczos ground states, Krylov $e^{-itH}\psi$                                                                          |
| <b>dynamics</b>  | exact Schrödinger/von Neumann evolution, RK4, Lindblad equation, $e^{t\mathcal{L}}$                                                                       |
| <b>optimize</b>  | parametrisations with prescribed contrast, batched multistart <code>extremize</code>                                                                      |
| <b>cost</b>      | run-time and memory estimates, warnings and errors                                                                                                        |

Table 3: Modules of `torch_vn_algebra`.

tors  $1/(\lambda_i - \lambda_j)$  and fail at degenerate spectra; the parametrisations of Section 6 avoid them. `Operator.eigenvalues()` computes spectra without eigenvectors, which is also 1.2–3.5 times faster than a full decomposition on a CPU.

**GPUs, tensor cores and TF32.** The library computes in FP32 or FP64 (real or complex); it does not use half precision. GPU generations differ strongly in what this costs. Tensor cores, absent on the P100 used for the first version, accelerate matrix products on Volta and later GPUs; from Ampere (A100) on, FP32 products may be executed in the TF32 format with a 10-bit mantissa, which is fast but accurate only to about  $10^{-3}$  relative. PyTorch keeps true FP32 for matrix products by default, and so does the library; the paper scripts expose a `-tf32` switch to measure the effect and record the setting. Eigen- and singular-value solvers do not use tensor cores. FP64 throughput is half of FP32 on data-centre GPUs (P100, V100, A100, H100), the latter two with FP64 tensor cores, but only 1/32–1/64 on consumer GPUs. We therefore recommend a data-centre GPU for double-precision work; the full-size runs of this paper are intended for an A100 or newer, and all generated tables state the hardware they were produced on.

## Number field and precision

Every algebra is created with two independent switches, `complex_valued` (real or complex matrices) and `precision` (`'single'` or `'double'`), giving the storage types `float32`, `float64`, `complex64` and `complex128`; everything built on the algebra (random operators, states, channels, chains, parametrisations) inherits them. `alg.like(complex_valued=..., precision=...)` creates the same sector structure with other numerics and `op.cast(alg2)` moves an operator there, e.g. to explore in single precision and re-evaluate the interesting cases in double precision.

**The field is part of the model.** Choosing real matrices is not only a matter of speed: the symmetry group and the random ensembles change. Haar matrices of a real algebra are orthogonal, so moments, level statistics and extremal values refer to  $O(n)$  instead of  $U(n)$  (e.g.  $\mathbb{E}|U_{11}|^4 = 3/(n(n+2))$  instead of  $2/(n(n+1))$ , Table 6); random real symmetric and complex Hermitian matrices have different level statistics, with mean spacing ratios 0.5321 and 0.6021 for GOE and GUE [17] (computed with the library). Real arithmetic is *exact* (loses nothing) when the problem is real: spectra and eigenvectors of real symmetric matrices, ground states and their entanglement for Hamiltonians without complex hoppings, statistics of time-reversal-invariant systems, and also some extremal problems — for real  $X, Y \geq 0$  the supremum of  $|\text{Tr}(XUY)|$  over unitaries is attained by a real orthogonal matrix

(the polar decomposition of the real matrix  $YX$  is real). Complex matrices are *required* for unitary time evolution  $e^{-iHt}$  and Lindblad dynamics with a Hamiltonian (the library refuses to run them in a real algebra), for Haar unitaries, CUE/CSE and  $SU(n)$ , for random pure states and quantities such as Page’s entropy, and for systems without time-reversal symmetry (fluxes, complex couplings).

**Precision.** Single precision resolves about  $10^{-7}$  relative to the largest entry. It is sufficient for Monte Carlo statistics whose statistical error is much larger (moments, histograms, spacing ratios with  $10^3$ – $10^5$  samples) and for the first, exploratory pass of an optimisation. Double precision is needed when the quantity of interest is small compared with the operators: an inequality checked near equality, small spectral gaps, entropies and logarithms of nearly singular states, contrasts close to 1, comparisons with exact results beyond six digits, long RK4 integrations and optimisation to high relative accuracy. Table 5 shows how drastic the difference can be: the von Neumann entropy of an almost pure state with eigenvalues of order  $10^{-9}$  is wrong by more than 100 % in single precision, and the round-off floor of a trace inequality that holds with equality is  $10^{-5}$  in single and  $10^{-13}$  in double precision.

**Cost.** An entry takes 4, 8, 8 and 16 bytes for `float32`, `float64`, `complex64` and `complex128`, so memory, and the batch that fits, changes by a factor up to four (`alg.bytes_per_operator(B)` reports it). A complex multiply–add costs about four real ones, and complex eigen- and singular-value solvers are two to four times slower than real ones. Double precision costs about a factor two on CPUs, but on GPUs it depends on the hardware: data-centre GPUs execute FP64 at half the FP32 rate, consumer GPUs at 1/32–1/64 of it, so that double precision there can be an order of magnitude slower. Table 4 gives the measured ratios for the current run; the recommendation that follows is to use the smallest type that is exact for the model (real if the problem is real) and the precision that the size of the effect requires.

| type                    | B/entry | MB/batch | generate | eigh f(A) | SVD  A | channel | expm | opt. step |
|-------------------------|---------|----------|----------|-----------|--------|---------|------|-----------|
| <code>float32</code>    | 4       | 1        | 1        | 1         | 1      | 1       | 1    | 1         |
| <code>float64</code>    | 8       | 2        | 1.5      | 1.5       | 1.1    | 1.7     | 1.7  | 1.2       |
| <code>complex64</code>  | 8       | 2        | 1.7      | 1.4       | 1.5    | 1.9     | 2    | 1.2       |
| <code>complex128</code> | 16      | 4        | 2.2      | 1.5       | 1.7    | 2.4     | 2    | 2.2       |

Table 4: Memory and time relative to `float32` for  $C = 8$  sectors of size  $k = 16$  and a batch of 128 operators (quick run on CPU (4 threads); `float32` times per sample: generate:  $105\ \mu\text{s}$ , eigh f(A):  $167\ \mu\text{s}$ , SVD |A|:  $298\ \mu\text{s}$ , channel:  $30.7\ \mu\text{s}$ , expm:  $57.4\ \mu\text{s}$ , opt. step:  $780\ \mu\text{s}$ ).

| type                    | $\max \ UU^* - 1\ $   | $\ \sqrt{A^2} - A\ /\ A\ $ | rel. error $S(\rho)$ , $\rho \approx \text{pure}$ | $ z $ at equality     |
|-------------------------|-----------------------|----------------------------|---------------------------------------------------|-----------------------|
| <code>float32</code>    | $8.3 \times 10^{-7}$  | $1.9 \times 10^{-6}$       | $2.3 \times 10^0$                                 | $2.3 \times 10^{-5}$  |
| <code>float64</code>    | $1.4 \times 10^{-15}$ | $4.0 \times 10^{-15}$      | $5.1 \times 10^{-9}$                              | $5.7 \times 10^{-14}$ |
| <code>complex64</code>  | $8.3 \times 10^{-7}$  | $2.2 \times 10^{-6}$       | $2.0 \times 10^0$                                 | $2.7 \times 10^{-5}$  |
| <code>complex128</code> | $1.1 \times 10^{-15}$ | $4.4 \times 10^{-15}$      | $5.0 \times 10^{-9}$                              | $5.0 \times 10^{-14}$ |

Table 5: Accuracy of the same computations in the four storage types: Haar unitaries ( $n = 64$ ), SVD square root, entropy of  $\rho = (1 - \varepsilon)|0\rangle\langle 0| + \varepsilon 1/n$  with  $\varepsilon = 10^{-7}$ , and the round-off floor of  $z = \text{Tr}(Y|X|Y) - \text{Tr}[YXY]$  for commuting  $X, Y$  (exactly 0).

**Safeguards.** Batched computations make it easy to request hours of work or more memory than the device has. The `cost` module therefore (a) times the first few items of every large batched eigen-, singular-value or QR decomposition and matrix exponential and issues a warning with the expected total time when it exceeds a threshold (10s by default), (b) compares the estimated peak memory of large allocations with the free memory of the device, warning above 50 % and raising an `InsufficientMemoryError` above 90 %, and (c) reports the expected duration of time-stepping and optimisation loops after their second step, optionally with a progress bar. All thresholds can be changed, and the checks can be switched off locally.

## 4 Usage by example

The listings of this section are complete programs from `examples/paper`; the test suite runs them, so they stay in sync with the code. They run on a CPU in seconds and use a GPU when available.

**Random operators with a prescribed contrast.** Listing 1 builds  $10^4$  positive operators in  $M_2 \oplus M_3 \oplus M_4$  whose spectra have contrast 0.6 and whose eigenvectors are Haar-random in every sector, and evaluates traces and spectral functions.

Listing 1: Random operators (`ex1_random_operators.py`).

```

"""Random operators with a prescribed spectrum in M = M_2 (+) M_3 (+) M_4."""
import torch
from torch_vn_algebra import TypeIAlgebra

dev = 'cuda' if torch.cuda.is_available() else 'cpu'
alg = TypeIAlgebra(n_factors=[2, 3, 4], k_factors=[2, 3, 4], device=dev)
B = 10_000                                     # Monte Carlo samples, processed as one batch

def contrast_sampler(delta):
    """Spectra in [lo, 1] with lambda_max = 1, lambda_min = lo, i.e. Michelson contrast delta."""
    lo = (1 - delta) / (1 + delta)

    def sampler(dim):
        lam = lo + (1 - lo) * torch.rand(B, dim, device=dev)
        lam[:, 0], lam[:, -1] = 1.0, lo
        return lam
    return sampler

# A_c = U_c diag(lambda) U_c^*, with independent Haar unitaries U_c in every block and sample
X = alg.operator_from_eigenvalues(contrast_sampler(0.6), batch_size=B,
                                force_positive=True, force_self_adjoint=True)

print(X.matrix.shape)                        # (10000, 3, 4, 4): batch, sector, padded block
print(X.michelson_contrast[:4])              # 0.6 for every sample
print(X.Tr_blunt().real.mean(), X.Tr_norm().real.mean(), X.tau_vN().real.mean())

# functional calculus through the spectral theorem, block by block
R = X.sqrt()
print((R @ R - X).frobenius_norm().max())    # ~1e-6 in single precision
print(X.apply_function(torch.log).tau_vN().real.mean()) # tau_vN(log X)

```

**States, measurement and trace conventions.** Listing 2 conditions random states on the outcome of a projective measurement (Lüders rule), which moves all probability into one sector, and

checks that expectation values do not depend on the trace used to represent the state.

Listing 2: States and conditioning (ex2\_states\_measurement.py).

```

"""States on M, sector probabilities, Lueders conditioning and trace conventions."""
import torch
from torch_vn_algebra import TypeIAlgebra, DensityMatrix, states

dev = 'cuda' if torch.cuda.is_available() else 'cpu'
alg = TypeIAlgebra([2, 3], [2, 3], device=dev)      # two superselection sectors

rho = states.random_density_matrix(alg, batch_size=4) # Hilbert-Schmidt random states
print(rho.sector_probabilities())                    # p_c = Tr rho_c, rows sum to 1

# measure the projection onto the first basis vector of sector 0
P = alg.from_blocks([torch.diag(torch.tensor([1.0, 0.0])), torch.zeros(3, 3)])
post, prob = rho.condition_on(P)                    # rho|P = P rho P / Tr(rho P)
print(prob, post.sector_probabilities())            # all weight now in sector 0

# the same state as a density w.r.t. tau_vN: omega(A) = tau_vN(d A) = Tr(rho A)
A = alg.operator_from_eigenvalues(lambda d: torch.randn(4, d), batch_size=4,
                                  force_self_adjoint=True)

d_tau = rho.density('tau_vN')
print(rho.expectation(A).real, (alg.operator(d_tau) @ A).tau_vN().real)
back = DensityMatrix.from_density(alg, d_tau, trace='tau_vN')
print((back.matrix - rho.matrix).abs().max())

```

**A map between sectors.** Two fermionic modes have sectors  $N = 0, 1, 2$  of dimensions 1, 2, 1. Particle loss with Kraus operators  $\sqrt{1-g}N$ ,  $\sqrt{g}a_1$ ,  $\sqrt{g}a_2$  maps sector  $N$  to  $N$  and  $N-1$ . Listing 3 builds it from its blocks; the classical part is the transition matrix of a death process,

$$T = \begin{pmatrix} 1 & g & 0 \\ 0 & 1-g & 2g \\ 0 & 0 & 1-2g \end{pmatrix},$$

which the program prints. The last lines illustrate a simple fact: a unital positive map cannot increase the Michelson contrast, since  $\lambda_{\min}1 \leq A \leq \lambda_{\max}1$  implies the same bounds for  $\Phi(A)$ ; the non-unital loss channel increases it for most samples.

Listing 3: Particle loss between sectors (ex3\_channels.py).

```

"""Particle loss between charge sectors of two fermionic modes, as an InterSectorChannel."""
import torch
from torch_vn_algebra import TypeIAlgebra, InterSectorChannel, channels, states

dev = 'cuda' if torch.cuda.is_available() else 'cpu'
# sectors N = 0, 1, 2 with bases |00>, |10>, |01>, |11>
alg = TypeIAlgebra([1, 2, 1], [1, 2, 1], device=dev)
g = 0.2                                     # loss probability per particle
s, z = g ** 0.5, torch.zeros
# Kraus operators K_0 = sqrt(1 - g)N, K_1 = sqrt(g) a_1, K_2 = sqrt(g) a_2, split into blocks (d, c)
blocks = {
    (0, 0): [torch.ones(1, 1)],
    (1, 1): [(1 - g) ** 0.5 * torch.eye(2)],
    (2, 2): [(1 - 2 * g) ** 0.5 * torch.ones(1, 1)],
    (0, 1): [z(1, 2), s * torch.tensor([[1.0, 0.0]]), s * torch.tensor([[0.0, 1.0]])],
    (1, 2): [z(2, 1), s * torch.tensor([[0.0], [1.0]]), s * torch.tensor([[1.0], [0.0]])],
}

```

```

loss = InterSectorChannel.from_blocks(alg, alg, blocks)
print(loss.is_trace_preserving())
print(loss.transition_matrix()[0])          # classical death process on N

rho = states.random_density_matrix(alg, batch_size=3)
print(loss(rho).sector_probabilities())     # still a DensityMatrix

# unital channels cannot increase the Michelson contrast, non-unital ones can
X = alg.operator_from_eigenvalues(lambda d: 0.5 + torch.rand(1000, d, device=dev),
                                batch_size=1000, force_positive=True)
mix = channels.random_mixed_unitary_channel(alg, 3, batch_size=1000)
print(bool((mix(X).michelson_contrast <= X.michelson_contrast + 1e-6).all()))
print((loss(X).michelson_contrast > X.michelson_contrast).float().mean())

```

**Open-system dynamics.** Listing 4 integrates spontaneous emission of a two-level atom with the RK4 solver and with the exact exponential of the Lindblad generator, and compares both with  $P_e(t) = P_e(0)e^{-\gamma t}$  and  $|\rho_{eg}(t)| = |\rho_{eg}(0)|e^{-\gamma t/2}$ .

Listing 4: Spontaneous emission (ex4\_dynamics.py).

```

"""Spontaneous emission of a two-level atom: Lindblad evolution against the exact solution."""
import math
import torch
from torch_vn_algebra import TypeIAlgebra, DensityMatrix, dynamics

dev = 'cuda' if torch.cuda.is_available() else 'cpu'
alg = TypeIAlgebra([2], [2], device=dev, precision='double')
omega, gamma = 1.0, 0.3
H = alg.from_blocks([torch.diag(torch.tensor([0.5 * omega, -0.5 * omega]))]) # basis |e>, |g>
L = alg.from_blocks([torch.tensor([[0.0, 0.0], [1.0, 0.0]])]) # sigma_- = |g><e|
psi = torch.tensor([0.6, 0.8], dtype=torch.complex128)
rho0 = DensityMatrix(alg, matrix=torch.outer(psi, psi.conj())[None, None].to(dev))

times = [0.0, 1.0, 2.0, 5.0]
rk4 = dynamics.lindblad_evolve(rho0, H, [L], times, rates=[gamma], substeps=100)
for t, rho in zip(times, rk4):
    exact = dynamics.lindblad_channel(alg, H, [L], t=t, rates=[gamma])(rho0) # exp(tL)
    p_e = 0.36 * math.exp(-gamma * t) # analytic excited population
    coh = 0.48 * math.exp(-gamma * t / 2) # analytic |rho_eg|
    m = rho.matrix[0, 0]
    print(f"t={t}: P_e {m[0, 0].real:.6f} (exact {p_e:.6f}), |rho_eg| {m[0, 1].abs():.6f} "
          f"(exact {coh:.6f}), RK4 vs exp(tL): {(rho.matrix - exact.matrix).abs().max():.1e}")

```

**Tensor products.** Listing 5 estimates the mean entanglement entropy of random pure states on  $\mathbb{C}^m \otimes \mathbb{C}^n$  and compares it with Page's formula [13].

Listing 5: Page entropy (ex5\_page\_entropy.py).

```

"""Average entanglement entropy of random pure states on C^m (x) C^n (Page, 1993)."""
import torch
from torch_vn_algebra import TypeIAlgebra, tensor_product, partial_trace, states

dev = 'cuda' if torch.cuda.is_available() else 'cpu'
m, n, B = 4, 8, 20_000
A = TypeIAlgebra([m], [m], device=dev, precision='double')
Bsys = TypeIAlgebra([n], [n], device=dev, precision='double')
AB = tensor_product(A, Bsys)

```

```

psi = states.random_density_matrix(AB, batch_size=B, rank=1)      # Haar-random pure states
S = states.von_neumann_entropy(partial_trace(psi, keep=1))        # entropy of rho_A

page = sum(1.0 / k for k in range(n + 1, m * n + 1)) - (m - 1) / (2 * n)
print(f"<S_A> = {S.mean():.5f} +- {S.std() / B ** 0.5:.5f}, Page: {page:.5f}")

```

**Searching for a violation.** Listing 6 compares Haar sampling with batched gradient ascent for  $\sup_U (|\text{Tr}(XUY)| - \text{Tr}(XY))$ , whose exact value  $\|YX\|_1 - \text{Tr}(XY)$  is positive for non-commuting  $X, Y \geq 0$ . Section 6 studies this systematically.

Listing 6: Sampling versus optimisation (ex6\_inequality\_search.py).

```

"""Searching for violations of |Tr(XUY)| <= Tr(XY): random sampling versus gradient ascent."""
import torch
from torch_vn_algebra import TypeIAlgebra

dev = 'cuda' if torch.cuda.is_available() else 'cpu'
torch.manual_seed(0)
k = 16
alg = TypeIAlgebra([k], [k], device=dev, precision='double')
pos = lambda d: 0.2 + torch.rand(1, d)
X = alg.operator_from_eigenvalues(pos, force_positive=True, force_self_adjoint=True)
Y = alg.operator_from_eigenvalues(pos, force_positive=True, force_self_adjoint=True)
trXY = (X @ Y).trace.real

def z(U):
    # z(U) = |Tr(XUY)| - Tr(XY), batched over U
    return (X @ U @ Y).trace.abs() - trXY

# 1) Monte Carlo over Haar unitaries
print("sampling, 10^5 Haar U:", z(alg.random_unitary_operator(100_000)).max().item())

# 2) gradient ascent over U = U0 exp(G - G^*), 64 random starts optimised in parallel;
# the exponential chart is re-centred at the current point every 50 steps
U0 = alg.random_unitary_operator(64).matrix
for _ in range(10):
    G = torch.zeros(64, 1, k, k, dtype=alg.hilbert.dtype, device=dev, requires_grad=True)
    opt = torch.optim.Adam([G], lr=0.05)
    for step in range(50):
        U = alg.operator(U0) @ alg.operator(G - G.conj().transpose(-2, -1)).expm()
        loss = -z(U).sum()
        opt.zero_grad()
        loss.backward()
        opt.step()
    U0 = U.matrix.detach()
print("gradient ascent:      ", z(alg.operator(U0)).max().item())

# exact supremum: sup_U |Tr(U YX)| = ||YX||_1
print("exact sup:          ", ((Y @ X).trace_norm() - trXY).item())

```

**Spin chains.** Listing 7 works with a Heisenberg ring of ten spins. A uniform field moves the ground state through the sectors until the chain is fully polarised above the saturation field  $h_s = 2J$ ; 100 realisations of random fields are diagonalised in one batch to compute the mean level-spacing ratio;

and the entanglement entropy of the ground state is computed from reduced states. Section 7 checks these tools against exact results.

Listing 7: Spin chains (ex7\_spin\_chain.py).

```
"""Spin chains with conserved S^z: sectors, disorder averages and entanglement."""
import torch
from torch_vn_algebra import SpinChain, DensityMatrix

dev = 'cuda' if torch.cuda.is_available() else 'cpu'
chain = SpinChain(L=10, boundary='periodic', device=dev)      # sectors N = 0..10, dims binom(10, N)
print(chain.algebra.k_factors)

# Heisenberg ring in a uniform field: the ground state changes sector as h grows
for h in (0.0, 1.0, 2.5):
    w = [blk[0] for blk in chain.heisenberg(h=h).eigenvalues()]
    print(f"h={h}: ground state in sector N={min(range(11), key=lambda N: w[N].min())}")

# random fields h_i ~ U[-W, W]: 100 disorder realisations in one batch, sector N = 5 only
for W in (0.5, 8.0):
    H = chain.random_field_heisenberg(W, batch_size=100, sector=5)
    E = H.eigenvalues()[0]                                     # (100, 252)
    r = SpinChain.level_spacing_ratio(E[:, 63:189]).mean()
    print(f"W={W}: <r> = {r:.3f} (GOE 0.531, Poisson 0.386)")

# entanglement entropy of the ground state of the clean chain, left block of n sites
E, V = chain.heisenberg(sector=5).eigh()[0]
psi = V[0, :, 0]
rho = DensityMatrix(chain.sector_algebra(5), matrix=torch.outer(psi, psi.conj())[None, None])
print([round(chain.entanglement_entropy(rho, n, sector=5).item(), 4) for n in range(1, 6)])
```

**Large sectors and open chains.** Dense matrices of a sector of dimension  $\binom{L}{L/2}$  stop fitting at about  $L = 16$ – $18$ . Listing 8 uses the sparse form of the Hamiltonian (a batch of diagonals and one shared hopping matrix) with Lanczos and Krylov solvers, splits a ring into momentum sectors, and lets particles leak out of the chain through Lindblad jumps between sectors.

Listing 8: Large sectors and open chains (ex8\_large\_and\_open\_chains.py).

```
"""Beyond dense matrices: Lanczos, Krylov evolution, momentum sectors and particle loss."""
import math
import torch
from torch_vn_algebra import SpinChain, krylov, dynamics

dev = 'cuda' if torch.cuda.is_available() else 'cpu'

# Lanczos in a sector of dimension binom(16, 8) = 12870, sparse Hamiltonian
ring = SpinChain(16, 'periodic', complex_valued=False, device=dev)
E0, psi, res = krylov.ground_state(ring.xx_sparse(sector=8))
print(f"Heisenberg ring L=16: E0 = {E0.item():.9f} (exact -7.142296361), residual {res.item():.1e}")

# Krylov evolution of a domain wall in the XX chain, L = 14 (sector dimension 3432)
chain = SpinChain(14, device=dev)
psi0 = chain.vector_in_sector('1' * 7 + '0' * 7)
psi_t = krylov.evolve(chain.xx_sparse(J=1.0, Delta=0.0, sector=7), psi0[None], [0.0, 3.0])
mag = (psi_t[-1, 0].abs() ** 2) @ chain.bits(7).to(torch.float64) - 0.5
print("S^z_i(t=3):", [round(x, 3) for x in mag.tolist()])
```

```

# translation symmetry: blocks of fixed momentum k = 2 pi m / L inside the sector N = 5 of a ring of
10
ring10 = SpinChain(10, 'periodic', device=dev)
Hk = ring10.xxz_momentum(J=1.0, Delta=1.0, sector=5)
print("momentum block sizes:", ring10.momentum_dims(5))
e0 = {m: w[0].min().item() for m, w in zip(Hk.algebra.charges, Hk.eigenvalues())}
print("ground state momentum m =", min(e0, key=e0.get), "(pi for L/2 odd)")

# particle loss S-i on every site with rate g: the sector distribution stays binomial
small = SpinChain(5, 'periodic', device=dev)
rho = dynamics.lindblad_evolve(small.basis_state('11011'), small.xxz(1.0, 0.7),
                               [small.lowering(i) for i in range(5)], [0.0, 1.0], rates=[0.4] * 5)

[-1]
q = math.exp(-0.4)
print("p_N(t=1):", [round(x, 6) for x in rho.sector_probabilities()[0].tolist()])
print("binomial:", [round(math.comb(4, n) * q ** n * (1 - q) ** (4 - n), 6) for n in range(5)] +
      [0.0])

```

**A noisy quantum circuit with a conserved quantity.** The quantum alternating operator ansatz solves constrained optimisation problems with “XY mixers”  $\sum (X_i X_j + Y_i Y_j)$  that keep the Hamming weight of the computational basis states, so the evolution stays in the feasible subspace [29]. In the language of this paper the qubit register carries the algebra  $\bigoplus_{N=0}^n M_{\binom{n}{N}}$  of Hamming-weight sectors, the mixer is an element of it, amplitude damping (time  $T_1$ ) is a Lindblad jump between sectors and pure dephasing (time  $T_\varphi$ , which together with  $T_1$  sets the coherence time  $T_2$ ,  $1/T_2 = 1/(2T_1) + 1/T_\varphi$ ) a jump inside them, and reading out the Hamming weight is the measurement of a central projection. Listing 9 evolves  $|111000\rangle$  (weight  $w = 3$ ) under the mixer for a time  $T = 2$  with both kinds of noise, at rates  $\gamma_1$  (damping) and  $\gamma_2$  (dephasing), and post-selects on the initial weight. With damping only, the post-selected fidelity is exactly one and the acceptance probability is  $e^{-w\gamma_1 T}$ : the no-jump evolution is generated by  $H - \frac{i}{2}\gamma_1 N$ , and  $N$  is central, so it only rescales the state. This exact statement is part of the test suite. Dephasing stays inside the sector and is not detected (fidelity with the noiseless state):

| $\gamma_1$ | $\gamma_2$ | $P(\text{weight } 3)$ | fidelity | after post-selection |
|------------|------------|-----------------------|----------|----------------------|
| 0.02       | 0          | 0.8869                | 0.8869   | 1.0000               |
| 0          | 0.02       | 1.0000                | 0.9730   | 0.9730               |
| 0.02       | 0.02       | 0.8869                | 0.8629   | 0.9730               |
| 0.1        | 0.1        | 0.5488                | 0.4805   | 0.8755               |

Listing 9: XY mixer with amplitude damping, dephasing and post-selection (ex9\_qaoa\_xy\_mixer.py).

```

"""
Quantum-computing example: an XY mixer (QAQA for constrained problems) on 6 qubits keeps the
Hamming weight, i.e. it acts inside one sector of M = (+)_N M_binom(6,N). Amplitude damping (T1)
moves weight to lower sectors, dephasing (T2) does not; measuring the Hamming weight - a central
projection of M - detects the first kind of error and post-selection restores part of the fidelity.
"""

import torch
from torch_vn_algebra import SpinChain, dynamics, states

dev = 'cuda' if torch.cuda.is_available() else 'cpu'
n, w, T = 6, 3, 2.0
qubits = SpinChain(n, 'periodic', device=dev)
weights
# qubits, Hamming weight, evolution time
# ring of qubits, sectors = Hamming

```

```

H = qubits.xxZ(J=1.0, Delta=0.0) # XY mixer sum (X_i X_j + Y_i Y_j) / 4
rho0 = qubits.basis_state('111000') # feasible initial state, weight 3
ideal = dynamics.von_neumann(rho0, H, [T])[0] # noiseless reference (pure)
P = qubits.algebra.central([1.0 if N == w else 0.0 for N in range(n + 1)]) # Hamming-weight projector

T1_jumps = [qubits.lowering(i) for i in range(n)] # |1> -> |0>: between sectors
T2_jumps = [qubits.sz(i) for i in range(n)] # dephasing: inside sectors
print(" gamma1 gamma2 P(weight 3) fidelity fidelity after post-selection")
for g1, g2 in ((0.0, 0.0), (0.02, 0.0), (0.0, 0.02), (0.02, 0.02), (0.1, 0.1)):
    rho = dynamics.lindblad_evolve(rho0, H, T1_jumps + T2_jumps, [0.0, T],
                                   rates=[g1] * n + [g2] * n, substeps=80)[-1]
    post, p_ok = rho.condition_on(P) # Lueders rule for the central projection
    F = states.trace_of_product(rho, ideal).real.item() # <psi_ideal| rho |psi_ideal>
    F_post = states.trace_of_product(post, ideal).real.item()
    print(f" {g1:5.2f} {g2:5.2f} {p_ok.item():10.4f} {F:8.4f} {F_post:8.4f}")

```

## 5 Validation against exact results

Table 6 compares Monte Carlo estimates produced with the library (double precision) with exact values: fourth moments of Haar unitaries and orthogonal matrices; the Weingarten-calculus value  $\mathbb{E} \tau((AUBU^*)^2) = -1/(N^2 - 1)$  for traceless involutions  $A = B$  and Haar  $U$ ; Page's formula for  $\langle S(\rho_A) \rangle$ ; and the mean purity  $(k + r)/(kr + 1)$  of induced random states of rank  $r$  [14]. The last column is the deviation in units of the standard error; all are of order one. For the circular ensembles the Wigner surmise is only an approximation to the exact spacing distribution, so no  $z$ -score is given; Figure 1 shows the distributions, with  $L^1$  distances 0.031, 0.021 and 0.029 for  $\beta = 1, 2, 4$  from the surmise; level repulsion  $P(s) \sim s^\beta$  is clearly visible. The spontaneous-emission example agrees with the analytic solution to  $2.2 \times 10^{-16}$  (exact exponential) and  $2.1 \times 10^{-8}$  (RK4).

**Trace functionals.** The three traces are checked on random non-Hermitian operators in  $M_2 \oplus M_3 \oplus M_5$  (Table 7): normalisation on the identity ( $\sum_c k_c$ ,  $C$  and 1), the tracial property  $\text{tr}(AB) = \text{tr}(BA)$ , invariance under unitaries of  $\mathcal{M}$ , and positivity  $\text{tr}(A^*A) > 0$  (faithfulness); the last column is the smallest ratio  $\text{tr}(A^*A)/\|A\|_F^2$  over the samples, which is bounded below by  $\min_c w_c$ . All three return values in the number field of the algebra (complex traces of non-Hermitian operators).

## 6 From sampling to optimisation

The first version of this work [1] studied three trace functionals by Monte Carlo:

$$\begin{aligned}
\text{Exp. 1: } z &= |\text{Tr}(XUY)| - \text{Tr}(XY), & X, Y \geq 0, U \in \mathcal{M} \text{ unitary,} \\
\text{Exp. 2: } z &= \text{Tr}|XY| - \text{Tr}(|X||Y|), & X = UX_0, Y = VY_0, X_0, Y_0 \geq 0, \\
\text{Exp. 3: } z &= \text{Tr}(Y|X|Y) - \text{Tr}|YXY|, & X = X^*, Y \geq 0,
\end{aligned}$$

with  $\text{Tr} = \text{Tr}_{\text{blunt}}$ , operators normalised to  $\lambda_{\max} = 1$ , and the contrasts of the operators as parameters. Two observations motivate a different approach.

**Sampling misses the extremes.** For fixed  $X, Y$  the supremum in Exp. 1 is known exactly,  $\sup_U z = \|YX\|_1 - \text{Tr}(XY) > 0$ . Table 8 and Figure 2 show that the best of 20 000 Haar samples is already negative for  $k \geq 4$  and moves further away from the supremum as  $k$  grows, while

| quantity                          | parameters       | exact       | Monte Carlo  | std. error           | $z$  |
|-----------------------------------|------------------|-------------|--------------|----------------------|------|
| $\mathbb{E} U_{11} ^4, U(n)$      | $n = 2$          | 0.333333    | 0.335015     | $1.5 \times 10^{-3}$ | +1.1 |
| $\mathbb{E} U_{11} ^2, U(n)$      | $n = 2$          | 0.5         | 0.5014       | $1.4 \times 10^{-3}$ | +1.0 |
| $\mathbb{E} U_{11} ^4, U(n)$      | $n = 4$          | 0.1         | 0.0999686    | $6.8 \times 10^{-4}$ | 0.0  |
| $\mathbb{E} U_{11} ^4, U(n)$      | $n = 8$          | 0.0277778   | 0.0277718    | $2.4 \times 10^{-4}$ | 0.0  |
| $\mathbb{E} U_{11} ^4, U(n)$      | $n = 16$         | 0.00735294  | 0.00737778   | $7.1 \times 10^{-5}$ | +0.4 |
| $\mathbb{E} U_{11} ^4, U(n)$      | $n = 32$         | 0.00189394  | 0.00188255   | $2.0 \times 10^{-5}$ | -0.6 |
| $\mathbb{E} U_{11} ^2, U(n)$      | $n = 32$         | 0.03125     | 0.031172     | $1.5 \times 10^{-4}$ | -0.5 |
| $\mathbb{E} U_{11} ^4, O(n)$      | $n = 2$          | 0.375       | 0.373641     | $1.8 \times 10^{-3}$ | -0.7 |
| $\mathbb{E} U_{11} ^2, O(n)$      | $n = 2$          | 0.5         | 0.498956     | $1.8 \times 10^{-3}$ | -0.6 |
| $\mathbb{E} U_{11} ^4, O(n)$      | $n = 4$          | 0.125       | 0.12365      | $9.8 \times 10^{-4}$ | -1.4 |
| $\mathbb{E} U_{11} ^4, O(n)$      | $n = 8$          | 0.0375      | 0.0378452    | $4.0 \times 10^{-4}$ | +0.9 |
| $\mathbb{E} U_{11} ^4, O(n)$      | $n = 16$         | 0.0104167   | 0.0104973    | $1.4 \times 10^{-4}$ | +0.6 |
| $\mathbb{E} U_{11} ^4, O(n)$      | $n = 32$         | 0.00275735  | 0.0027189    | $3.9 \times 10^{-5}$ | -1.0 |
| $\mathbb{E} U_{11} ^2, O(n)$      | $n = 32$         | 0.03125     | 0.0311461    | $2.1 \times 10^{-4}$ | -0.5 |
| $\mathbb{E} \tau((AUBU^*)^2)$     | $N = 4$          | -0.0666667  | -0.0668132   | $1.8 \times 10^{-3}$ | -0.1 |
| $\mathbb{E} \tau((AUBU^*)^2)$     | $N = 16$         | -0.00392157 | -0.00342506  | $4.0 \times 10^{-4}$ | +1.3 |
| $\mathbb{E} \tau((AUBU^*)^2)$     | $N = 64$         | -0.0002442  | -0.000192229 | $2.0 \times 10^{-4}$ | +0.3 |
| $\langle s^2 \rangle, \text{COE}$ | $N = 32$         | 1.27324     | 1.28476      | $5.2 \times 10^{-3}$ | -    |
| $\langle s^2 \rangle, \text{CUE}$ | $N = 32$         | 1.1781      | 1.17783      | $3.8 \times 10^{-3}$ | -    |
| $\langle s^2 \rangle, \text{CSE}$ | $N = 32$         | 1.10447     | 1.10321      | $3.9 \times 10^{-3}$ | -    |
| $\langle S(\rho_A) \rangle$       | $m = 2, n = 8$   | 0.600372    | 0.600235     | $9.6 \times 10^{-4}$ | -0.1 |
| $\langle S(\rho_A) \rangle$       | $m = 4, n = 8$   | 1.15314     | 1.15389      | $1.0 \times 10^{-3}$ | +0.7 |
| $\langle S(\rho_A) \rangle$       | $m = 8, n = 8$   | 1.58853     | 1.58838      | $8.7 \times 10^{-4}$ | -0.2 |
| $\mathbb{E} \text{Tr} \rho^2$     | $k = 4, r = 2$   | 0.666667    | 0.666228     | $1.0 \times 10^{-3}$ | -0.4 |
| $\mathbb{E} \text{Tr} \rho^2$     | $k = 4, r = 4$   | 0.470588    | 0.470432     | $6.8 \times 10^{-4}$ | -0.2 |
| $\mathbb{E} \text{Tr} \rho^2$     | $k = 16, r = 4$  | 0.307692    | 0.307736     | $2.0 \times 10^{-4}$ | +0.2 |
| $\mathbb{E} \text{Tr} \rho^2$     | $k = 16, r = 16$ | 0.124514    | 0.124468     | $5.4 \times 10^{-5}$ | -0.8 |

Table 6: Monte Carlo estimates versus exact values (quick run on CPU (4 threads), PyTorch 2.14.1+cu130).

batched gradient ascent (32 starts, 500 steps) finds it to a relative accuracy of about  $10^{-5}$  or better (median  $10^{-6}$  or better). This is concentration of measure: the unitaries close to the maximiser form a set whose Haar measure decays rapidly with the dimension. In particular, the absence of violations in Monte Carlo samples for  $k = 16$  reported in [1] is a property of the sampling, not of the inequality.

**Constrained optimisation.** The natural objects are the envelopes

$$\bar{z}(\delta) = \sup\{z : \Delta(X) = \delta\}, \quad \underline{z}(\delta) = \inf\{z : \Delta(X) = \delta\},$$

(and the same with  $\Delta(Y)$ , or with both contrasts prescribed), the supremum being taken over all remaining variables. The module `optimize` parametrises the constraint set exactly: the spectrum is  $\lambda_{\max} = 1$ ,  $\lambda_{\min} = (1 - \delta)/(1 + \delta)$  and the other eigenvalues in between (through a sigmoid), the eigenvectors are  $U = U_0 e^{G - G^*}$  with a Haar-random  $U_0$  per start and a chart that is re-centred after every round, and for Exp. 3 the signs of the eigenvalues of  $X$  are drawn per start and kept fixed. Every grid point and every start is one element of the batch, so the whole envelope is computed in a single batched optimisation.

Figures 3 and 4 and Table 9 show the result. Every Monte Carlo sample of [1] lies between the computed envelopes; samples that fall outside the linear interpolation between grid points are re-optimised at their exact contrast, after which none remains outside. For a single sector of size

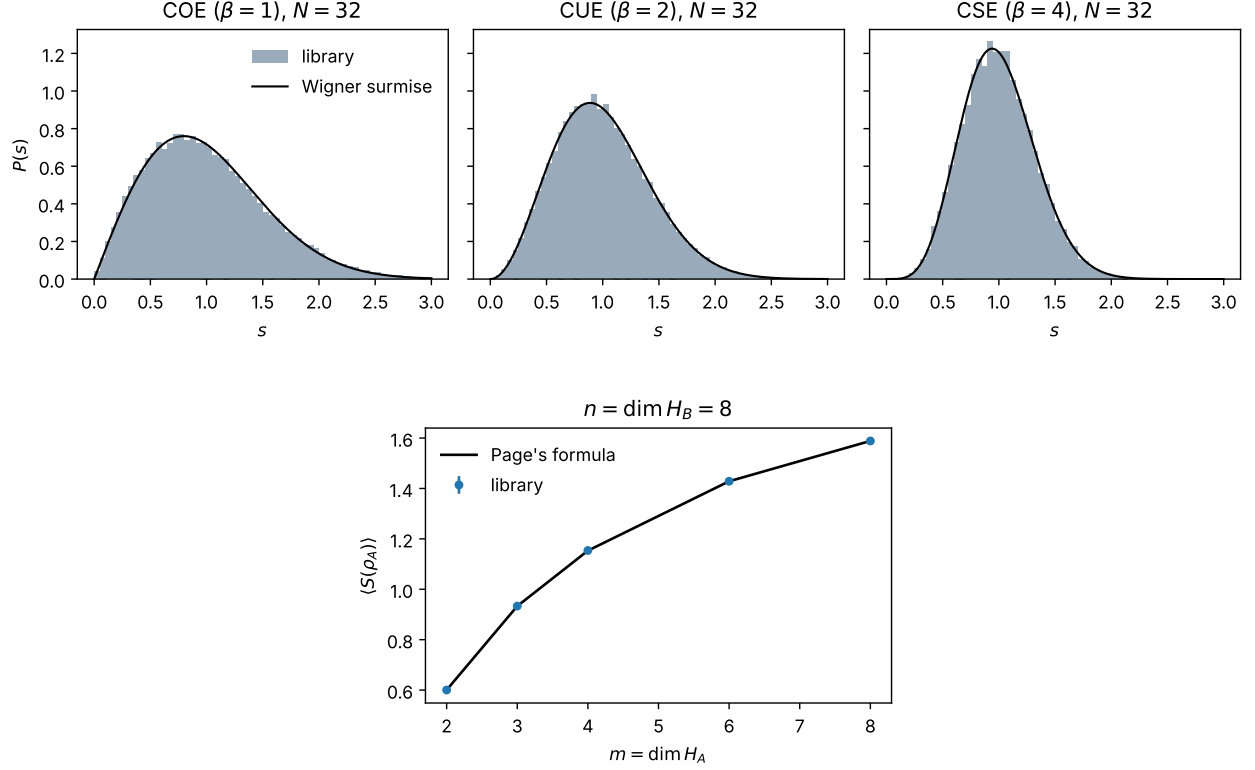


Figure 1: Top: nearest-neighbour spacings of COE, CUE and CSE eigenphases generated by the library, against the Wigner surmise. Bottom: mean entanglement entropy of random pure states against Page’s formula (error bars: two standard errors).

| trace                      | error of $\text{tr}(1)$ | $\max  \text{tr}(AB) - \text{tr}(BA) $ | $\max  \text{tr}(UAU^*) - \text{tr}(A) $ | $\min \text{tr}(A^*A)/\ A\ _F^2$ |
|----------------------------|-------------------------|----------------------------------------|------------------------------------------|----------------------------------|
| $\text{Tr}_{\text{blunt}}$ | 0                       | $4.0 \times 10^{-15}$                  | $7.1 \times 10^{-15}$                    | 1.000                            |
| $\text{Tr}_{\text{norm}}$  | 0                       | $1.8 \times 10^{-15}$                  | $1.9 \times 10^{-15}$                    | 0.223                            |
| $\tau_{\text{vN}}$         | 0                       | $4.5 \times 10^{-16}$                  | $6.3 \times 10^{-16}$                    | 0.074                            |

Table 7: Identities of the three trace functionals, 2000 random operators in double precision.

$k = 2$  the samples come within a few per cent of the envelopes; with two sectors they already miss a substantial part of the range (for Exp. 1,  $\sup z$  is about 1.5 times the largest sampled value), and for larger  $k$  sampling misses the extremes entirely (Table 8).

Two envelopes can be checked analytically. In Exp. 3,  $z \geq 0$  always:

**Proposition 1.** *For  $X = X^*$  and  $Y \geq 0$ ,  $\text{Tr} |YXY| \leq \text{Tr}(Y|X|Y)$ , with equality if  $Y$  commutes with  $X$ .*

*Proof.* With  $X = X_+ - X_-$ ,  $|X| = X_+ + X_-$ , the triangle inequality for the trace norm gives  $\|YX_+Y - YX_-Y\|_1 \leq \text{Tr}(YX_+Y) + \text{Tr}(YX_-Y)$ . If  $XY = YX$ , then  $|YXY| = Y^2|X|$ .  $\square$

The computed lower envelope of Exp. 3 is zero to machine precision. In Exp. 1, the lower envelope for prescribed  $\Delta(X) = \delta$  equals  $-(kC - 2\delta/(1 + \delta))$  when all sectors have size  $k \geq 2$ :  $\text{Tr}(XY) \leq \text{Tr} X \leq kC - 1 + \lambda_{\min}(X)$  with equality for  $Y = 1$ , while  $|\text{Tr}(UYX)|$  can be made zero by a cyclic permutation of the eigenbasis of  $X_c$  in every sector. The optimiser reproduces this curve to a relative deviation

| $k$ | violation found |        | $\sup z / \text{Tr } XY$ | best sample<br>$z / \text{Tr } XY$ | ascent rel. error    |                      | time [s] |        |
|-----|-----------------|--------|--------------------------|------------------------------------|----------------------|----------------------|----------|--------|
|     | sampling        | ascent |                          |                                    | median               | max                  | sampl.   | ascent |
| 2   | 2/4             | 4/4    | $4.4 \times 10^{-5}$     | $-4.1 \times 10^{-4}$              | $5.4 \times 10^{-8}$ | $1.7 \times 10^{-5}$ | 0.018    | 1      |
| 4   | 0/4             | 4/4    | $2.0 \times 10^{-3}$     | -0.19                              | $9.4 \times 10^{-8}$ | $7.5 \times 10^{-7}$ | 0.063    | 1.2    |
| 8   | 0/4             | 4/4    | $3.2 \times 10^{-3}$     | -0.58                              | $5.0 \times 10^{-7}$ | $2.1 \times 10^{-6}$ | 0.18     | 1.3    |
| 16  | 0/4             | 4/4    | $4.5 \times 10^{-3}$     | -0.78                              | $2.6 \times 10^{-6}$ | $4.4 \times 10^{-6}$ | 0.76     | 2.4    |

Table 8: Exp. 1 for fixed random  $X, Y$ : Haar sampling versus gradient ascent (medians over 4 pairs; quick run on CPU (4 threads)).

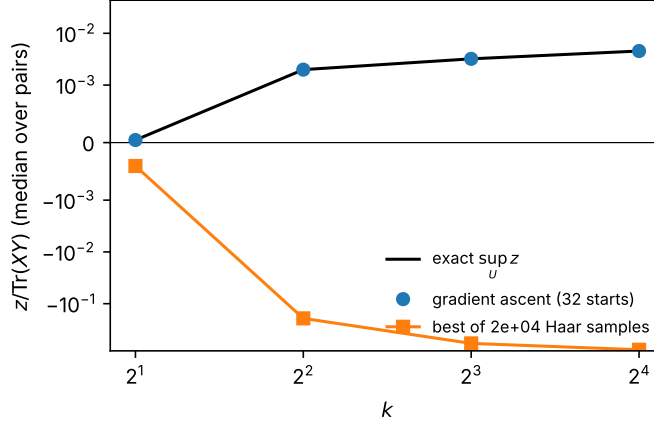


Figure 2: Best  $z$  found by sampling and by gradient ascent, relative to  $\text{Tr}(XY)$ , against the exact supremum (symmetric log scale).

of  $8.1\text{e-}03$ . The upper envelopes of all three experiments are, to our knowledge, not known in closed form; the computed curves are lower bounds for them (a local search) and suggest conjectures for sharp contrast-dependent inequalities, which we leave for future work.

## 7 Symmetric tensor products and spin chains

The basis of `SpinChain` is, by construction, that of the iterated fused product of single sites: the tests check that  $S_i^z$  built by the chain coincides with  $S_i^z \otimes 1$  formed with `kron` in the fused product of the algebra of `SpinChain(L-1)` with a single site. Beyond that, integrable chains provide exact results for every part of the module (Table 11, Figure 5; script `scripts/paper/chains.py`):

- *Free fermions.* By the Jordan–Wigner transformation the open XX chain ( $\Delta = 0$ ) is a free-fermion model with single-particle energies  $\varepsilon_k = J \cos \frac{\pi k}{L+1}$ , so the spectrum of sector  $N$  consists of all sums of  $N$  distinct  $\varepsilon_k$ . For  $L = 10$  all  $2^L$  levels agree to  $8.0 \times 10^{-15}$ .
- *Heisenberg ring.* Ground-state energies agree with the tabulated exact values to the nine digits of the table;  $E_0/L$  approaches the Bethe-ansatz value  $\frac{1}{4} - \ln 2 \approx -0.4431$ . Every level of sector  $N < L/2$  reappears in sector  $N + 1$  ( $\text{SU}(2)$  multiplets), up to  $1.5 \times 10^{-14}$ .
- *Dynamics.* The melting of a domain wall  $|\uparrow \cdots \uparrow \downarrow \cdots \downarrow\rangle$  in the XX chain, evolved in the sector  $N = L/2$  (dimension 924 for  $L = 12$ ), reproduces the single-particle prediction  $\langle n_i(t) \rangle =$

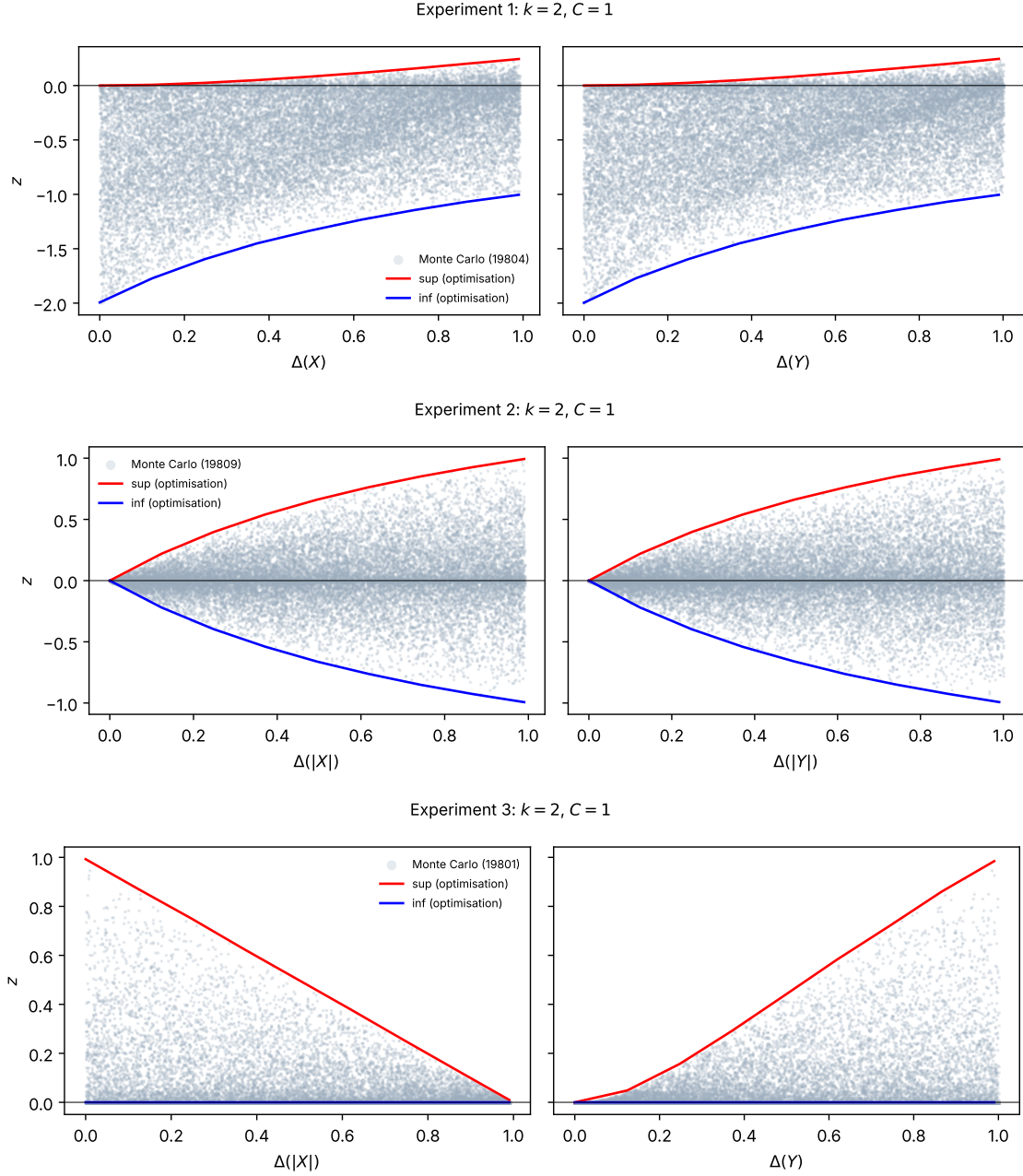


Figure 3: Sharp bounds versus sampling for the three trace functionals ( $k = 2, C = 1$ ; one row per experiment). Horizontal axis: Michelson contrast of the first operator (left) or of the second operator (right); vertical axis:  $z$  for operators normalised to  $\lambda_{\max} = 1$ . Grey: the Monte Carlo samples of [1]; red and blue: the upper and lower envelopes found by constrained optimisation, the other contrast being free. The envelopes enclose every sample; in this smallest case the samples come close to them, which is no longer so for two sectors or larger blocks (Tables 8 and 9). In Exp. 1 the region  $z > 0$  (violation of  $|\text{Tr}(XUY)| \leq \text{Tr}(XY)$ ) extends to  $z \approx 0.24$ ; in Exp. 3 the lower envelope is the proven bound  $z \geq 0$ .

| Exp. | $k$ | $C$ | inf $z$ | sup $z$ | Monte Carlo range of $z$ | outside |
|------|-----|-----|---------|---------|--------------------------|---------|
| 1    | 2   | 1   | -1.994  | +0.2436 | $[-1.98, +0.225]$        | 0       |
| 1    | 2   | 2   | -3.991  | +0.4765 | $[-3.39, +0.328]$        | 0       |
| 1    | 4   | 1   | -3.994  | +0.472  | —                        | —       |
| 2    | 2   | 1   | -0.9931 | +0.9934 | $[-0.921, +0.975]$       | 0       |
| 2    | 2   | 2   | -1.956  | +1.963  | $[-1.43, +1.72]$         | 0       |
| 2    | 4   | 1   | -1.961  | +1.957  | —                        | —       |
| 3    | 2   | 1   | 0       | +0.9927 | $[0, +0.946]$            | 0       |
| 3    | 2   | 2   | 0       | +1.968  | $[0, +1.39]$             | 0       |
| 3    | 4   | 1   | 0       | +1.962  | —                        | —       |

Table 9: Global extremes of  $z$  over all contrasts found by constrained optimisation, the range of the Monte Carlo samples, and the number of samples outside the envelopes after re-optimisation (12 starts per grid point, 240 steps; quick run on CPU (4 threads)).

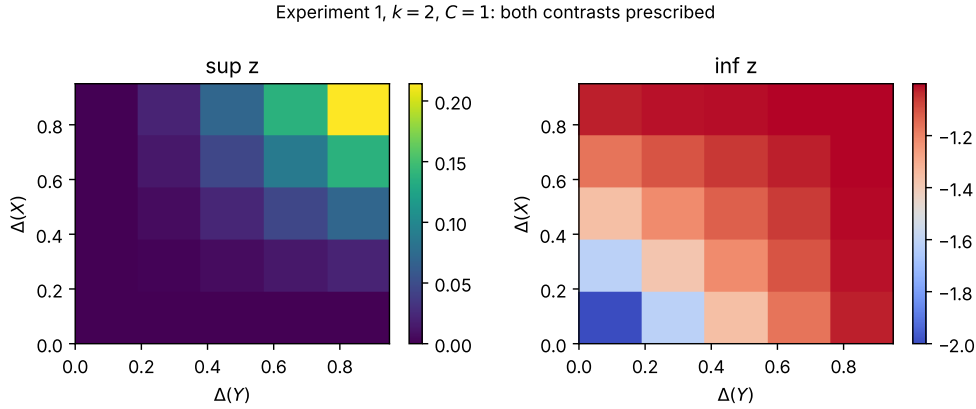


Figure 4: Exp. 1,  $k = 2$ ,  $C = 1$ , both contrasts prescribed: sup  $z$  (left) and inf  $z$  (right) on a grid of  $\Delta(Y)$  (horizontal) and  $\Delta(X)$  (vertical). The violation sup  $z > 0$  needs both operators to be far from scalar and grows with both contrasts.

$[e^{iht}C(0)e^{-iht}]_{ii}$ , with  $h$  the single-particle hopping matrix and  $C_{ij} = \langle c_i^\dagger c_j \rangle$  the correlation matrix, to  $5.0 \times 10^{-15}$ .

- *Entanglement.* The entropy of a block of  $n$  sites of the XX ground state, obtained with **reduced\_state**, agrees with Peschel's formula  $S = -\sum_\nu [\nu \ln \nu + (1 - \nu) \ln(1 - \nu)]$  over the eigenvalues  $\nu$  of the restricted correlation matrix [15], including the parity oscillations of open chains, to  $3.3 \times 10^{-12}$ .
- *Disorder.* For the Heisenberg chain with random fields  $h_i \in [-W, W]$  the mean ratio of consecutive level spacings in the middle of the spectrum crosses over from the GOE value 0.5307 to the Poisson value  $2 \ln 2 - 1 \approx 0.3863$  [16, 17] as  $W$  grows, the hallmark of many-body localisation. Here the batch dimension holds the disorder realisations (40 per point).
- *Momentum.* For periodic chains the basis  $|a, k\rangle = N_a^{-1/2} \sum_j e^{-ikj} T^j |a\rangle$  over orbit representatives  $a$  of the translation  $T$  splits sector  $N$  into blocks of fixed momentum  $k = 2\pi m/L$  (for  $L = 12$  and  $N = L/2$  of sizes 80, 75, 78, 76, 78, 75, 80, 75, 78, 76, 78, 75); the union of their spectra is the spectrum of the sector to  $3.8 \times 10^{-14}$ , and the ground state of the Heisenberg ring has  $k = 0$  for even and  $k = \pi$  for odd  $L/2$  (Marshall's sign rule). Momentum blocks are

complex even for real Hamiltonians, one of the cases of Section 3 where the field is fixed by the problem.

- *Beyond dense matrices.* Lanczos with full re-orthogonalisation and restarts, batched over Hamiltonians, reproduces the free-fermion ground-state energies of open XX chains and the Heisenberg ring in sectors far beyond dense diagonalisation (Table 10); Krylov evolution of the domain wall for  $L = 16$  (sector dimension 12870) agrees with the single-particle solution to  $1.1 \times 10^{-14}$ .
- *Open chains.* With jump operators  $S_i^-$  on every site (Lindblad terms between sectors  $N \rightarrow N - 1$ ) and any  $N$ -conserving Hamiltonian the particle number performs a pure death process, so the sector distribution must be binomial,  $p_n(t) = \binom{N_0}{n} e^{-n\gamma t} (1 - e^{-\gamma t})^{N_0 - n}$ ; the library reproduces it to  $2.4 \times 10^{-10}$ .

| model                     | sector dim | $E_0$ (Lanczos) | $E_0$ (exact) | difference            | time [s] |
|---------------------------|------------|-----------------|---------------|-----------------------|----------|
| XX open, $L = 12$         | 924        | -3.6481149053   | -3.6481149053 | $8.9 \times 10^{-16}$ | 0.034    |
| XX open, $L = 16$         | 12870      | -4.9189757237   | -4.9189757237 | $3.6 \times 10^{-15}$ | 0.31     |
| XX open, $L = 18$         | 48620      | -5.5547827927   | -5.5547827927 | $8.9 \times 10^{-16}$ | 1.4      |
| Heisenberg ring, $L = 16$ | 12870      | -7.1422963606   | -7.1422963610 | $3.8 \times 10^{-10}$ | 0.38     |

Table 10: Batched Lanczos on sparse sector Hamiltonians: ground-state energies against free fermions ( $\sum$  of the  $L/2$  lowest  $\cos \frac{\pi k}{L+1}$ ) and the tabulated Heisenberg value (nine digits); quick run on CPU (4 threads).

| $L$ | dim ( $N = L/2$ ) | $E_0$ (library) | $E_0$ (exact) | difference            | $E_0/L$  |
|-----|-------------------|-----------------|---------------|-----------------------|----------|
| 4   | 6                 | -2.000000000    | -2.000000000  | $1.0 \times 10^{-16}$ | -0.50000 |
| 6   | 20                | -2.802775638    | -2.802775638  | $2.7 \times 10^{-10}$ | -0.46713 |
| 8   | 70                | -3.651093409    | -3.651093409  | $6.3 \times 10^{-11}$ | -0.45639 |
| 10  | 252               | -4.515446354    | -4.515446354  | $4.9 \times 10^{-10}$ | -0.45154 |
| 12  | 924               | -5.387390917    | -5.387390917  | $4.5 \times 10^{-10}$ | -0.44895 |

Table 11: Ground-state energy of the Heisenberg ring  $H = \sum_i \mathbf{S}_i \cdot \mathbf{S}_{i+1}$ ; exact values to nine digits (quick run on CPU (4 threads)).

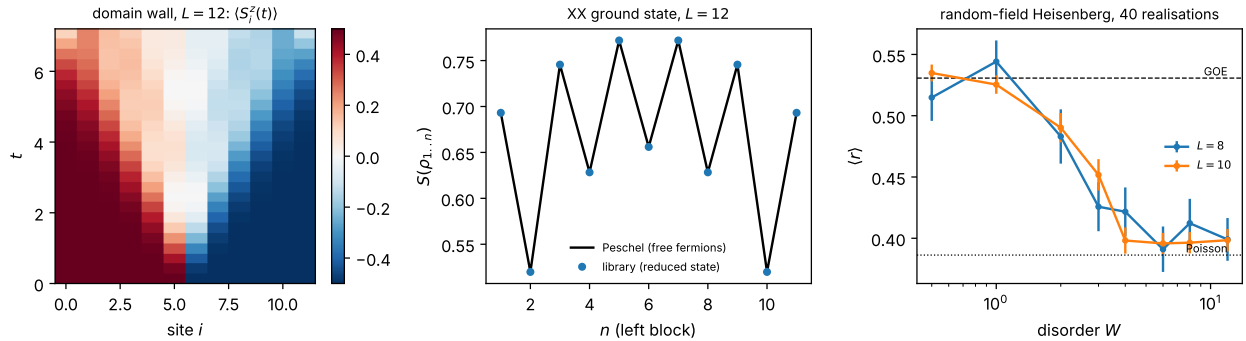


Figure 5: Left:  $\langle S_i^z(t) \rangle$  after a domain-wall quench of the XX chain (light cone). Middle: entanglement entropy of the XX ground state, library vs. Peschel's formula. Right: mean level-spacing ratio of the random-field Heisenberg chain, with the GOE and Poisson values.

## 8 Comparison with NumPy/SciPy

To separate what the library adds from what plain array code already gives, the same computations were written three ways (`scripts/paper/baselines.py`; Table 12). (A) The Monte Carlo of Exp. 1:  $B$  pairs of positive operators with prescribed contrast and Haar-orthogonal eigenvectors in every sector, a Haar-orthogonal  $U \in \mathcal{M}$ , and  $z$ . The direct NumPy/SciPy formulation loops over samples and sectors with `scipy.stats.ortho_group`; a hand-vectorised NumPy version uses stacked QR decompositions with the phase correction; the library needs three calls. (B) Ground states of sector Hamiltonians: `scipy.sparse.linalg.eigsh` (ARPACK [36]), one call per Hamiltonian, against the batched Lanczos solver. The three Monte Carlo versions agree within their statistical errors, and the ground-state energies agree to round-off.

The comparison is instructive in both directions. On a CPU the gain over the loop depends on the block size: it is large for small blocks, where the Python overhead of the loop dominates (about two orders of magnitude for  $k = 2$ ), and small for large blocks, where both spend their time in the same LAPACK routines (a factor of about two for  $k = 16$ ,  $C = 8$ ). Against carefully vectorised NumPy the library is only about 1.5 times faster: its value there is that the vectorised, correctly sampled, block-aware version is the default rather than an expert rewrite, and that the same code runs on a GPU and can be differentiated. On a CPU, ARPACK is faster than our Lanczos solver (which re-orthogonalises fully for robustness): by a factor of about three for one Hamiltonian and of under two for a batch of ten. Batching narrows the gap and is what lets the solver run on a GPU, but on a CPU we recommend SciPy for sparse eigenproblems.

| $C$ | $k$ | NumPy loop | NumPy vectorised | library (CPU) | library (GPU) |
|-----|-----|------------|------------------|---------------|---------------|
| 1   | 2   | 104        | 1.79             | 1.14          | –             |
| 8   | 16  | 1.02e+03   | 713              | 448           | –             |

| $L$ | sector dim | Hamiltonians | <code>eigsh</code> [s] | library [s] | $\max  \Delta E_0 $ |
|-----|------------|--------------|------------------------|-------------|---------------------|
| 14  | 3432       | 1            | 0.00741                | 0.024       | 4.4e-15             |
| 14  | 3432       | 10           | 0.075                  | 0.125       | 2.5e-14             |
| 16  | 12870      | 1            | 0.0274                 | 0.0476      | 8.9e-16             |

Table 12: Top: time per Monte Carlo sample [ $\mu$ s] for (A); bottom: ground states of Heisenberg rings (one Hamiltonian) and random-field chains ( $W = 3$ , several Hamiltonians) in the sector  $N = L/2$  (quick run on CPU (4 threads)).

## 9 Performance

Table 13 and Figure 6 report the wall-clock time per sample of library operations (not of isolated kernels), in single precision, with the batch size chosen so that the batch holds about  $2^{22}$  matrix entries (quick run on CPU (4 threads); CPU with 4 threads). Expected behaviour: for single small blocks the CPU is faster, because kernel launches dominate; the GPU gains for many sectors and for the matrix products in channels and in the optimisation step; batched symmetric eigen- and singular-value solvers lose much of their advantage from  $k \approx 64$  on, where the GPU libraries switch from batched small-matrix kernels to general ones.

The first version of this work reported GPU speedups for raw kernels on a P100 and found them only for part of the configurations. That observation reflects where GPUs help: a GPU needs enough independent work per kernel launch. For computations in (1) this means many samples, many sectors

or large blocks, and matrix products rather than decompositions of small matrices. The design follows from it: everything is batched, so that the batch dimension supplies the parallel work, and the cost module warns before configurations that would be slow. For workloads that are small or dominated by a single eigenproblem, the CPU (or SciPy, Section 8) is the right tool, and the library runs there unchanged.

| operation | $k$ | $C$ | batch | time per sample [ $\mu$ s] |
|-----------|-----|-----|-------|----------------------------|
| generate  | 4   | 1   | 4096  | 1.33                       |
| f(A) eigh | 4   | 1   | 4096  | 3.19                       |
| A  SVD    | 4   | 1   | 4096  | 6.14                       |
| contrast  | 4   | 1   | 4096  | 2.07                       |
| channel   | 4   | 1   | 4096  | 0.595                      |
| exp(tL)   | 4   | 1   | 4096  | 59.6                       |
| opt. step | 4   | 1   | 4096  | 13.6                       |
| generate  | 4   | 16  | 1024  | 28.1                       |
| f(A) eigh | 4   | 16  | 1024  | 44.1                       |
| A  SVD    | 4   | 16  | 1024  | 90                         |
| contrast  | 4   | 16  | 1024  | 24.3                       |
| channel   | 4   | 16  | 1024  | 7.81                       |
| exp(tL)   | 4   | 16  | 1024  | 1.42e+03                   |
| opt. step | 4   | 16  | 1024  | 314                        |
| generate  | 16  | 1   | 1024  | 18.1                       |
| f(A) eigh | 16  | 1   | 1024  | 35.8                       |
| A  SVD    | 16  | 1   | 1024  | 67.2                       |
| contrast  | 16  | 1   | 1024  | 14                         |
| channel   | 16  | 1   | 1024  | 2.96                       |
| exp(tL)   | 16  | 1   | 1024  | 1.67e+04                   |
| opt. step | 16  | 1   | 1024  | 205                        |
| generate  | 16  | 16  | 64    | 439                        |
| f(A) eigh | 16  | 16  | 64    | 495                        |
| A  SVD    | 16  | 16  | 64    | 938                        |
| contrast  | 16  | 16  | 64    | 200                        |
| channel   | 16  | 16  | 64    | 46.7                       |
| exp(tL)   | 16  | 16  | 64    | 2.48e+05                   |
| opt. step | 16  | 16  | 64    | 3e+03                      |

Table 13: Wall-clock time per sample of library operations for  $C$  sectors of size  $k$  and the batch size shown (single precision; “exp( $tL$ )” uses a smaller batch because its superoperators have  $k^4$  entries). With a GPU the table also gives the CPU time and the speedup CPU/GPU.

## 10 Limitations and roadmap

- *Layout.* Blocks are padded to  $k_{\max}$ . For very different sector sizes (binomial sectors of spin chains) one works in single sectors; operators acting on all sectors at once would need a bucketed layout, which is not implemented yet.
- *Symmetries.* Abelian charges with an arbitrary fusion rule and lattice translations are supported; non-abelian symmetries (SU(2) multiplets, Clebsch–Gordan structure), reflections and spin flip are not used to reduce blocks further. Dense methods reach about 16–18 spins, the sparse Lanczos and Krylov solvers should reach about 24–26 spins per sector on a GPU with

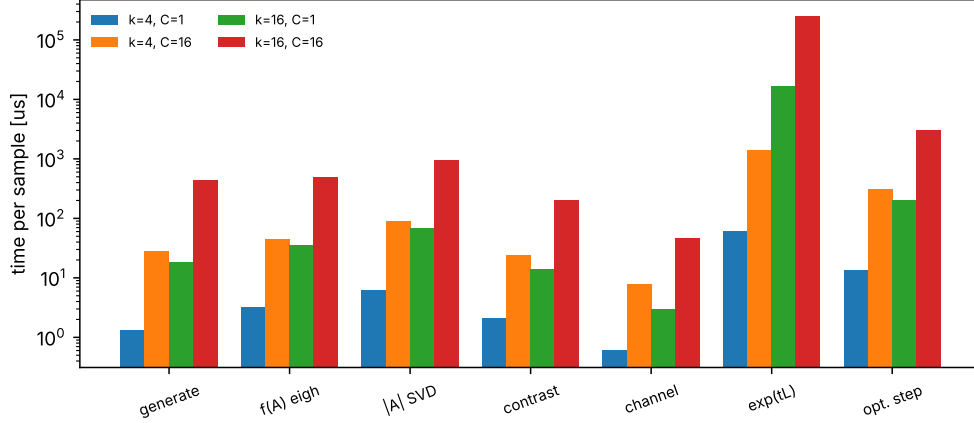


Figure 6: Time per sample (logarithmic axis) of the operations of Table 13, one bar per configuration  $(k, C)$ .

tens of gigabytes (an estimate from the memory of the Krylov basis, not a measurement); tensor-network methods are out of scope.

- *Optimisation.* The search is local; envelopes are lower bounds for suprema (upper bounds for infima). Gradients propagated through eigen- or singular vectors fail at degenerate spectra; the parametrisations avoid this for the contrast constraint, but not for every user-defined objective.
- *Scale.* Choi matrices, superoperators and  $e^{t\mathcal{L}}$  work with  $k_c^2 \times k_c^2$  matrices and are limited to blocks of a few tens; RK4 integration scales to larger blocks.
- *Scope.* Finite dimensions and Type I only; no tensor-network methods, and sparse methods only for sector Hamiltonians (Lanczos and Krylov).

## 11 Availability and reproducibility

The code (MIT licence) is available at [https://github.com/andre-a-no/von\\_neumann\\_type\\_i](https://github.com/andre-a-no/von_neumann_type_i). `bash scripts/paper/run_all.sh full cuda` regenerates every table and figure of this text on a GPU; `bash scripts/paper/run_all.sh` runs a reduced version on a CPU in a few minutes. The tables state on which device and in which mode they were produced.

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